

MSE 311 Thermodynamics of materials-Homework 11

Name:

Student ID:

Please hand in this homework before class on December 1st.

1. Does the liquid atom have coordination numbers? Please show how to calculate it.

2. Prove the following equation when $\eta \neq 0$:

$$\Delta S^{M,id} > \Delta S^M, \text{ where } \Delta S^M = -\frac{R}{2} \left(\left(X_A + \frac{\eta}{2} \right) \ln \left(X_A + \frac{\eta}{2} \right) + \left(X_B + \frac{\eta}{2} \right) \ln \left(X_B + \frac{\eta}{2} \right) + \left(X_A - \frac{\eta}{2} \right) \ln \left(X_A - \frac{\eta}{2} \right) + \left(X_B - \frac{\eta}{2} \right) \ln \left(X_B - \frac{\eta}{2} \right) \right)$$

$$\Delta S^{M,id} - \Delta S^M$$

$$= -R(X_A \ln X_A + X_B \ln X_B) + \frac{R}{2} \left(\left(X_A + \frac{\eta}{2} \right) \ln \left(X_A + \frac{\eta}{2} \right) + \left(X_B + \frac{\eta}{2} \right) \ln \left(X_B + \frac{\eta}{2} \right) + \left(X_A - \frac{\eta}{2} \right) \ln \left(X_A - \frac{\eta}{2} \right) + \left(X_B - \frac{\eta}{2} \right) \ln \left(X_B - \frac{\eta}{2} \right) \right)$$

$$= \frac{R}{2} \left[\left(X_A + \frac{\eta}{2} \right) \ln \left(X_A + \frac{\eta}{2} \right) + \left(X_A - \frac{\eta}{2} \right) \ln \left(X_A - \frac{\eta}{2} \right) - 2X_A \ln X_A + \left(X_B + \frac{\eta}{2} \right) \ln \left(X_B + \frac{\eta}{2} \right) + \left(X_B - \frac{\eta}{2} \right) \ln \left(X_B - \frac{\eta}{2} \right) - 2X_B \ln X_B \right]$$

$$(x + \Delta x) \ln(x + \Delta x) - x \ln x + (x - \Delta x) \ln(x - \Delta x) - x \ln x$$

$$= \Delta x \frac{(x + \Delta x) \ln(x + \Delta x) - x \ln x}{\Delta x} - \Delta x \frac{(x - \Delta x) \ln(x - \Delta x) - x \ln x}{-\Delta x}$$

$$= \Delta x \left(x \ln x \right)' \Big|_{x=x+\frac{\Delta x}{2}} - \Delta x \left(x \ln x \right)' \Big|_{x=x-\frac{\Delta x}{2}}$$

$$= \Delta x^2 \frac{\Delta x \left(x \ln x \right)' \Big|_{x=x+\frac{\Delta x}{2}} - \Delta x \left(x \ln x \right)' \Big|_{x=x-\frac{\Delta x}{2}}}{\Delta x}$$

$$= \Delta x^2 \left(x \ln x \right)''$$

$$= \Delta x^2 \left(\ln x + 1 \right)'$$

$$= \Delta x^2 \frac{1}{x} > 0$$

$$\therefore \Delta S^{M,id} - \Delta S^M > 0$$

$$\Delta S^{M,id} > \Delta S^M$$

3. Copper and gold form the complete range of solid solution at temperatures between 410°C and 889°C, and, at 600°C, the excess molar Gibbs free energy of formation of the solid solutions is given by

$$G^{XS} = -28,280 X_{Au} X_{Cu} \text{ J}$$

Calculate the partial pressures of Au and Cu exerted by the solid solution of $X_{Cu} = 0.6$ at 600°C.

The saturated vapor pressure of solid copper is given by

$$\ln p_{Cu}^{\circ} (\text{atm}) = -\frac{40,920}{T} - 0.86 \ln T + 21.67$$

and the saturated vapor pressure of solid gold is given by

$$\ln p_{Au}^{\circ} (\text{atm}) = -\frac{45,650}{T} - 0.306 \ln T + 10.81$$

Solution to Quantitative Problem 1

The solid solutions are regular with $\alpha = -28,280 \text{ J}$. Therefore, from Equation 9.64,

$$\ln \gamma_{Cu} = \frac{\alpha}{RT} X_{Au}^2 = -\frac{28,280 \times 0.4^2}{8.3144 \times 873} = -0.624$$

Thus,

$$\gamma_{Cu} = 0.536 \text{ and } a_{Cu} = \gamma_{Cu} X_{Cu} = 0.536 \times 0.6 = 0.322$$

Similarly,

$$\ln \gamma_{Au} = \frac{28,280 \times 0.6^2}{8.3144 \times 873} = -1.403$$

Thus,

$$\gamma_{Au} = 0.246 \text{ and } a_{Au} = 0.246 \times 0.4 = 0.098$$

Therefore, at 873 K,

$$p_{Cu}^{\circ} = 3.35 \times 10^{-14} \text{ atm}$$

and

$$p_{\text{Au}}^{\circ} = 1.52 \times 10^{-16} \text{ atm}$$

From Equation 9.12, $a_i = p_i/p_i^0$, and thus, the partial pressures exerted by the alloy are

$$p_{\text{Cu}} = 0.322 \times 3.35 \times 10^{-14} = 1.08 \times 10^{-14} \text{ atm}$$

and

$$p_{\text{Au}} = 0.098 \times 1.52 \times 10^{-16} = 1.50 \times 10^{-16} \text{ atm}$$

$$p_{\text{Au}}^{\circ} = 1.23 \times 10^{-19} \text{ atm}$$

$$p_{\text{Au}} = 1.23 \times 10^{-19} \times 0.098 = 1.20 \times 10^{-20} \text{ atm}$$

4. Please derive the following formula

$$X_{A(l)} = \frac{\left[1 - \exp\left(-\Delta G_{m(B)}^{\circ} / RT\right)\right] \exp\left(-\Delta G_{m(A)}^{\circ} / RT\right)}{\exp\left(-\Delta G_{m(A)}^{\circ} / RT\right) - \exp\left(-\Delta G_{m(B)}^{\circ} / RT\right)}$$

$$X_{A(s)} = \frac{1 - \exp\left(-\Delta G_{m(B)}^{\circ} / RT\right)}{\exp\left(-\Delta G_{m(A)}^{\circ} / RT\right) - \exp\left(-\Delta G_{m(B)}^{\circ} / RT\right)}$$